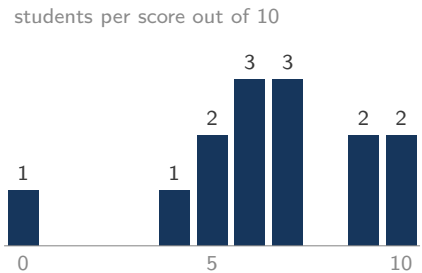
Quiz on Problem Set 3 — how the class did

14

students

6.6 /10 mean

median 6.75



The rent rises by EUR 100 a day; the ATC at the same output

one change

Suppose the workshop's rent rose by €100 a day and nothing else changed. At the output of question 3 (its efficient scale), what would its ATC be?



bars: the class's share of the points

Right answer: variant A: 8.96 · variant B: 6.80 · variant C: 6.64 (euros a bar)

Most often: a wrong number (5 of 14) or blank (4). The rent is a fixed cost: at the same output the ATC rises by 100 divided by that output (1 euro at 100 bars, 0.80 at 125).

A box costs EUR 10 more; the monopoly's new price and quantity

one change

Suppose a box cost the company €10 more to make than in your copy (the demand table stays the same). Which price in the table would it set now, and how many boxes would it sell?



bars: the class's share of the points

Right answer: variant A: price if a box costs 10 more 35, quantity if a box costs 10 more 2 · variant B: price if a box costs 10 more 30, quantity if a box costs 10 more 3 · variant C: price if a box costs 10 more 38, quantity if a box costs 10 more 3

Most often: blank (4 of 14) or a wrong number (4). Sell while marginal revenue is above the NEW cost (old cost + 10), then read the price for that quantity in the demand table.

Cim keeps a promise of a high price; what each shop earns

one change

Cim promises Vall that it will set a high price. Suppose Cim keeps its promise and Vall sets the price that earns Vall the most this month. What does each shop earn, in thousand euros a month?



bars: the class's share of the points

Right answer: variant A: Cim if Cim keeps a high price 7, Vall if Cim keeps a high price 16 · variant B: Cim if Cim keeps a high price 8, Vall if Cim keeps a high price 22 · variant C: Cim if Cim keeps a high price 11, Vall if Cim keeps a high price 18

Most often: blank (3 of 14) or another payoff. With Cim's price high, Vall earns most with a low price: Cim gets its high-against-low payoff, Vall its low-against-high payoff.

The hotel offers EUR 40 thousand; the economic profit

one change

This year the hotel offers Carme her job back at €40 thousand a year (instead of her old pay). If all else were as last year, what would her economic profit be this year?

the class (1 pt)  57%

bars: the class's share of the points

Right answer: variant A: -10 · variant B: -3 · variant C: -18 (thousand euros a year)

Most often: a wrong number (5 of 14). The implicit costs are now the new pay of 40 plus the interest on her savings: subtract both from the accounting profit.

Demand doubles; the number of farms in the long run

one change

Suppose that, at every price, buyers wanted twice as many boxes as in part (d), and nothing else changed. In the long run, how many farms would there be in all?



bars: the class's share of the points

Right answer: variant A: 220 · variant B: 320 · variant C: 270 (farms)

Most often: blank (3 of 14) or a wrong number (3). With constant costs the price stays at the lowest ATC and each farm keeps its output, so twice the demand needs twice the farms.

Carme's economic profit

from your solutions

Carme's economic profit last year, in thousand euros (a loss with a minus sign).



bars: the class's share of the points

Right answer: variant A: -7 · variant B: 10 · variant C: -9 (thousand euros a year)

Most often: mostly right. Economic profit = accounting profit minus the implicit costs (her old pay and the interest on her savings).

The efficient scale and its ATC

from your solutions

The workshop's efficient scale (the output where the average total cost, ATC, is lowest), and the ATC there.



bars: the class's share of the points

Right answer: variant A: efficient scale 100, ATC there 7.96 · variant B: efficient scale 125, ATC there 6 · variant C: efficient scale 100, ATC there 5.64

Most often: mostly right. The efficient scale is the output where the average total cost is lowest.

The monopoly's price and quantity

from your solutions

The price the company sets, and how many boxes it sells.



bars: the class's share of the points

Right answer: variant A: price 30, quantity 3 · variant B: price 26, quantity 4 · variant C: price 32, quantity 4

Most often: mostly right. The monopoly sells while marginal revenue is above marginal cost and charges the demand table's price for that quantity.

The long-run price and the number of farms

from your solutions

In the long run: the price of a box of artichokes, and the number of farms in all.



bars: the class's share of the points

Right answer: variant A: long-run price 18, farms in the long run 110 · variant B: long-run price 32, farms in the long run 160 · variant C: long-run price 22, farms in the long run 135

Most often: mostly right. In the long run the price is the lowest ATC; the number of farms is the quantity bought at that price divided by one farm's output.

What each shop earns in the outcome

from your solutions

What each shop will earn when both choose their prices as in part (b), in thousand euros a month.



bars: the class's share of the points

Right answer: variant A: Cim 10, Vall 7 · variant B: Cim 11, Vall 14 · variant C: Cim 14, Vall 12

Most often: mostly right. Both shops choose the price that is best whatever the other does.